# Ernest Skrzypczyk

*Bachelor of Science in Electrical Engineering - Control Engineering and Robotics*

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*France*
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*https://github.com/em-er-es/personal*

## Education

2015–current **Double Masters Degree**, *European Masters in Advanced Robotics + ( EMARO+)*, University of Genoa, Italy / Ecolé Centrale de Nantes, France, A programme of Erasmus Mundus.

2007–2011 **Bachelor of Science - Electrical Engineer - Control Engineering and Robotics**, *Wrocław University of Technology*, Wybrzeże Wyspiańskiego 27, PL-50370 Wrocław, *very good / bardzo dobry*.
ISCED 5A, Polish grading system; See supplement to diploma, page 5

2004–2006 **Abitur**, *Berufsbildende Schulen Meppen - gew. und kaufm. Fachrichtungen - Fachrichtung Technik*, Nagelhof 83, D-49716 Meppen,
*3,0 - good / gut.*
ISCED 3A, German grading system

## Master thesis

Title *UGV and UAV collaboration within an autonomous infrastructure*

Supervisor Docent / Maître de conférences Olivier Kermorgant

Description Development of a collaborative, decentralized and heterogeneous SLAM based on previous work within ROS and V-REP framework at LS2N, formelly known as IRCCYN.

Grade pending

## Engineering project

Title *Metrological properties of a hybrid amplifier*

Supervisor PhD / dr inż. Daniel Dusza

Description Design, construction and measurements of a hybrid amplifier with a working frequency within the audible range and a microcontroller based control system.

Grade excellent / celujący

Prizes and Awards — Stowarzyszenie Elektryków Polskich - Association of Polish Electrical Engineers
1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011

## Experience

### Vocational

01.02.2017 – current, ends 31.08.2017 — **Master thesis work**, *École Centrale de Nantes, LS2N (formely known as IRCCYN) – ARMEN*, F-44300 Nantes, France.
Work on the master thesis as part of the ARMEN team.

2011 – 2015 — **Self development**, *Improvement of skills in various areas, mostly GNU/Linux related*.

12.07.2010 – 20.08.2010 — **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein Berßen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems. Internship for studies at the Wrocław University of Technology.

14.07.2008 – 15.08.2008 — **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein Berßen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems.

2 weeks in May 2005 — **Computer scientist**, *Ebert Mikrokomputer GmbH*, D-49716 Meppen, Germany.
Computer system installations mostly in the private sector. Customer care on site. Secondary school internship.

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### EMARO+

UNIGE — **Projects and assignments:**

- *Localization of a biped humanoid robot using EKF within ROS framework (**Github repository**)*
- *Study and implementation of developed SLAM algorithms for a biped humanoid robot within ROS framework (**Github repository**)*
- *Mobile robots control: point and posture tracking, Lyapunov control, design and simulation*
- *Mobile robots motion planning using bug, distance based and probabilistic roadmap planning algorithms*
- *Offline and online tuning of EKF in a real world case scenario using landmarks*
- *Technical report on Agriculture in controlled environments*
- *Technical paper review and presentation on FLEXA*
- *Delmia simulations*
- *Design and evaluation of a user interface for a motion analysing application*
- *Basic image processing including edge and object detection, colour based segmentation, optical flow (Lucas-Kanade algorithm) and stereo vision*
- *POSIX compliant publisher / subscriber communication models using pipes and pthreads*
- *Realtime scheduling with RM, PS and EDF*

## Languages

Polish — **Native**

| | | |
|---|---|---|
| English | **B2+** | *Mark: very good / bardzo dobry at SJO WUT* |
| German | **C1** | *100% exam test score, 2nd place in contest at FLD WrUT* |
| Spanish | **A2** | *High school secondary language for 2 years* |
| Italian | **B1** | *Score 25/30, spent 1 year in Genova, Italy within EMARO+ programme* |
| French | **A1** | *About to spent 1 year in Nantes, France within EMARO+ programme* |

## Computer skills

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|---|---|---|---|
| Operating systems | Extensive experience with Microsoft Windows, GNU/Linux, some experience with hardware specific solutions and FreeBSD | Distributions | Arch Linux (current main OS), OpenSUSE, Funtoo, Gentoo, Linux Mint, Ubuntu, Rasbian |
| Software experience | Various CAD/CAE, office and scientific applications and suites across most media fields including audio, video, graphics, text, design, simulation and engineering | Applications | LibreOffice, Microsoft Office, Altium Designer, Adams, Simatic Step 7, CX-Programmer, Delmia, Matlab, Spyder, Pythonxy, V-REP, Scilab, Proto, Photoshop, Gimp, Inkscape, Ableton Live, Audacity, Geany, TeXstudio, nano |
| Shells / Command line interfaces | Zsh (current main shell), Bash, PowerShell, Command Prompt | Software and community contributions | Proto, Arch Linux Wiki articles (ALSA - Basic Syntax), Arch Linux Forums and packages maintenance, wine, q4wine |

## Repositories

| | | | |
|---|---|---|---|
| GitHub | http://github.com/em-er-es | Scripts | http://github.com/em-er-es/scripts |
| Robotics | http://github.com/em-er-es/rollo<br>http://github.com/em-er-es/rollo-slam | EMARO+ | http://github.com/em-er-es/reports |

## Programming languages

**General languages**

- *Assembler on various hardware*
- *BASIC on Commodore 64 and various BASIC derivatives on robots*
- *C/C++ in courses, assignments, projects and contributions*
- *Python*
- *LUA*

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**Frameworks**

- *ROS - Robot Operating System*
- *V-REP*

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**PLC languages**

- *Function block diagram*
- *Ladder logic*

Scripting
- *Bash / Zsh*
- *Matlab / Scilab / Pythonxy / Spyder*
- *Command Prompt / Batch*
- *Autohotkey*

## Certificates, Licences

Electrical qualification — **Operation position** in the area of: **operation**, **maintenance**, **repairs** and **assembly**

*SEP - AoPEE, Wrocław*
*Date of exam: 19.05.2010*
*Valid for 5 years*

Driving licence — **Class A, B**

*Driving licence acquired in Germany in 2005*

## Organisations, associations and memberships

SNS — Academic student association
STRIMER — Wrocław University of Technology

*Took part in several projects, educational trip and made a publication for the SSC 2009*

## Additional courses

Warsaw

2015 ABC of enterneurship

2015 Self presentation and interview skills

## References

Please contact me for specific information

## Publications

Rzepecki Bartłomiej, Skrzypczyk Ernest, *"Plasma and its applications"*, 7th Students' Science Conference, Wrocław, Poland, 2009

## Prizes and Awards

Prize — 1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011

Prize — Second place in German language tournament at FLD WrUT

Commendation — For best paper presentation at SSC 2009

Aknowledgment — Third place for best group work at SSC 2009

## Scholarships

Several scholarships for achievements in studies at the WrUT Faculty of EE

## Interests

**Arts**
- *Graphics, photography, image rendering and manipulation*
- *Music, audio creation and manipulation*
- *Paper model building*

**Philosophy**
- *New socio-economic structures*
- *Consequences of implementation of new technologies*

**Technology**
- *Self-sustainable autonomous vehicles, facilities and infrastructure*
- *Robot interaction*
- *Artificial intelligence*
- *FOSS, especially GNU/Linux*
- *Open Hardware, Open Technology, Open Economy*

## Additional information

**Mobility** Full mobility within 157 countries, especially across Europe

**Self-development** Main focus was to shift working environment to and towards FOSS, contribute and exchange with the FOSS community.